Chapter 16

Non-conforming Methods

Assume we are given a variational problem in some Hilbert space H, for example to find $u \in V \subset H$ such that

$$a(u,v) = l(v), \qquad \forall v \in V, \tag{16.1}$$

where $a : H \times H \to \mathbb{R}$ is a bilinear form, and $l : H \to \mathbb{R}$ is a linear functional. Standard examples are the elliptic problems discussed in previous chapters. (See, for example, Model Problem 4, where $H \equiv H^1(\Omega)$ and $V \equiv H_0^1(\Omega)$.)

We have previously seen how to formulate a Ritz-Galerkin approximation, using a finite-dimensional subspace $S_h \subset V$. For example, refer to Chapter 12 for general formulation, and Chapter 13 for analysis using piecewise linear approximation on triangular meshes, where h is a characteristic mesh size. This has been our standard setup so far. In an actual implementation we often violate the condition $S_h \subset V$, or use a bilinear from $a_h(\cdot,\cdot)$ in the approximation, which is distinct from the bilinear form $a(\cdot,\cdot)$ used in the variational formulation (16.1). This can happen for a variety of reasons. For example, we may use numerical quadrature instead of exact integration. To motivate, consider the Poisson problem with homogeneous boundary conditions, where $V \equiv H_0^1(\Omega)$ for some domain Ω, and

$$\begin{aligned} a(u,v) &= \int_\Omega (\nabla v)^T A \nabla u \, dx, \\ l(v) &= \int_\Omega f v \, dx. \end{aligned} \tag{16.2}$$

We assume $A = (a_{ij})_{i,j=1,\ldots,d}$ is symmetric and uniformly elliptic, and that the a_{ij} are some bounded functions on Ω. Let $u_h \in S_h \subset H_0^1(\Omega)$ solve the Ritz-Galerkin approximation

$$a(u_h, v_h) = l(v_h), \qquad \forall v_h \in S_h. \tag{16.3}$$

We have always tacitly assumed that the integration evaluating the bilinear form is exact. In an actual implementation, one might replace such integrals with a numerical quadrature, to allow one implementation to work with many different coefficient functions. Does this compromise accuracy?

Most quadrature approximations to (16.2) can be written

$$a_h(u_h, v_h) := \sum_{T \in \mathcal{T}_h} \sum_{q=1}^{N_q} w_q \nabla v_h(x_q) A(x_q) \nabla u_h(x_q) |T|, \tag{16.4}$$

where x_q, w_q are a set of quadrature nodes and weights, respectively, and $|T|$ is the area of the triangle T. We have thus introduced a bilinear from a_h, which is clearly mesh dependent, and distinct from $a(\cdot,\cdot)$ given in (16.2). We replace (16.3) with

$$a_h(u_h, v_h) = l_h(v_h) \qquad \forall v_h \in S_h, \ h > 0, \tag{16.5}$$

where a_h is defined in (16.4), and a similar approximation may be used for l_h.

For now, we shall still assume that $S_h \subset V$ for all $h > 0$. In general we define $a_h : S_h \times S_h \to \mathbb{R}$ as a mesh-dependent bilinear form on the discrete approximation space. (It need not be defined for elements in H.) It turns out that we can get a reasonable characterization of the error, provided the new mesh-dependent bilinear form is uniformly S_h-elliptic, i.e. there is a constant $\overline{\alpha} > 0$, independent of h, such that

$$a_h(v_h, v_h) \geq \overline{\alpha} \, \|v_h\|_H^2 \qquad \forall v_h \in S_h, \; h > 0. \tag{16.6}$$

We have the following theorem, which holds not only for the Poisson problem, but also for more general problems that may be formulated in (subsets of) a Hilbert space H:

Theorem 16.1 *Let H be a Hilbert space, and let $u \in V \subset H$ satsify*

$$a(u, v) = l(v), \qquad \forall v \in V, \tag{16.7}$$

where $a : H \times H \to \mathbb{R}$ is a continuous bilinear form, i.e. there is a constant C such that

$$a(u, v) \leq C \, \|u\|_H \, \|v\|_H \, , \qquad \forall u, v \in V.$$

Let $u_h \in S_h \subset V$ satisfy

$$a_h(u_h, v_h) = l_h(v_h), \qquad \forall v_h \in S_h. \tag{16.8}$$

Assume that $a_h(\cdot, \cdot)$ is uniformly S_h-elliptic, i.e. eq. (16.6) holds. Then

$$\|u - u_h\|_H \leq \inf_{\substack{v_h \in S_h \\ v_h \neq 0}} \left\{ \left(1 + \frac{C}{\overline{\alpha}}\right) \|u - v_h\|_H + \frac{1}{\overline{\alpha}} \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \left(\frac{|a(v_h, w_h) - a_h(v_h, w_h)|}{\|w_h\|_H} + \frac{|l(w_h) - l_h(w_h)|}{\|w_h\|_H} \right) \right\} \tag{16.9}$$

Proof Define $w_h := u_h - v_h$ for arbitrary $v_h \in S_h$, such that $v_h \neq u_h$. From coercivity (eq. (16.6)), we obtain

$$\begin{aligned} \overline{\alpha} \, \|w_h\|_H^2 &\leq a_h(w_h, w_h) \\ &= a_h(u_h - v_h, w_h) \pm a(u - v_h, w_h) \\ &= a(u - v_h, w_h) + (a(v_h, w_h) - a_h(v_h, w_h)) + (l_h(w_h) - l(w_h)) \end{aligned}$$

where we have used (16.7), and eq. (16.8) to replace $a(u, w_h)$ and $a_h(u_h, w_h)$ with the respective right-hand side. (Note: In the former case this works, because we still assume that $S_h \subset V$.) From continuity there follows

$$\overline{\alpha} \, \|w_h\|_H^2 \leq C \, \|u - v_h\|_H \, \|w_h\|_H + |a(v_h, w_h) - a_h(v_h, w_h)| + |l_h(w_h) - l(w_h)| \, .$$

Dividing by $\|w_h\|_H$ on both sides we have

$$\overline{\alpha} \, \|w_h\|_H \leq C \, \|u - v_h\|_H + \frac{|a(v_h, w_h) - a_h(v_h, w_h)|}{\|w_h\|_H} + \frac{|l_h(w_h) - l(w_h)|}{\|w_h\|_H}.$$

To get rid of the dependence of w_h on the right-hand side, we take the supremum to obtain

$$\overline{\alpha} \, \|w_h\|_H \leq C \, \|u - v_h\|_H + \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|a(v_h, w_h) - a_h(v_h, w_h)|}{\|w_h\|_H} + \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|l_h(w_h) - l(w_h)|}{\|w_h\|_H}.$$

By triangle inequality, there holds for any $v_h \in S_h$

$$\|u - u_h\| \leq \|u - v_h\| + \|v_h - u_h\| \, . \tag{16.10}$$

We finally obtain

$$\|u-u_h\|_H \le \left(1+\frac{C}{\overline{\alpha}}\right)\|u-v_h\|_H + \frac{1}{\overline{\alpha}}\left(\sup_{\substack{w_h\in S_h\\ w_h\neq 0}} \frac{|a(v_h,w_h)-a_h(v_h,w_h)|}{\|w_h\|_H} + \frac{|l_h(w_h)-l(w_h)|}{\|w_h\|_H}\right).$$

As v_h is arbitrary, we can take the infimum on the right-hand side, and the assertion follows. ■

The first term in eq. (16.9) of a nature similar to that proved in Theorem 12.2 for the standard Ritz-Galerkin approximation. It relates the error to the best approximation in S_h. The remaining terms can be thought of as additional errors that are incurred by replacing a with a_h, and likewise l with l_h. These errors vanish for the Ritz-Galerkin approximation. For the case of numerical quadrature, we conclude that we must choose the quadrature formula accurate enough, so that the additional error is of higher order compared to the best approximation error (which we can bound by the interpolation error proved in Theorem 13.4.)

Note carefully that in Theorem 16.1, we have only assumed that a_h is defined for elements in S_h. (In the proof of Theorem 16.1, the bilinear form is never evaluated with elements from H.) If, however, the mesh-dependent bilinear form makes sense for elements in H, and furthermore, we have

$$a_h(u,v_h) \le C\,\|u\|_H\,\|v_h\|_H\,, \qquad \forall u\in H,\ v_h\in S_h,$$

then the expression (16.9) simplifies somewhat. Indeed, we can write for $w_h = u_h - v_h$, where u_h solves (16.8), and $v_h \in S_h$ is arbitrary,

$$\begin{aligned}\overline{\alpha}\,\|w_h\|_H^2 &\le a_h(w_h,w_h)\\ &= a_h(u_h-v_h,w_h) \pm a_h(u,w_h)\\ &= a_h(u-v_h,w_h) + (l_h(w_h)-a_h(u,w_h))\\ &\le C\,\|u-v_h\|_H\,\|w_h\|_H + |l_h(w_h)-a_h(u,w_h)|\end{aligned}$$

Together with the triangle inequality (16.10) this yields

$$\|u_h-u\|_H \le \inf_{\substack{v_h\in S_h\\ v_h\neq 0}}\left(1+\frac{C}{\overline{\alpha}}\right)\|u-v_h\|_H + \frac{1}{\overline{\alpha}}\sup_{\substack{w_h\in S_h\\ w_h\neq 0}}\frac{|l_h(w_h)-a_h(u,w_h)|}{\|w_h\|_H}. \tag{16.11}$$

Now we dispense with the requirement that $S_h \subset V$. Let us again consider the variational problem of finding $u\in V$, such that

$$a(u,v) = l(v), \qquad \forall v\in V\subset H. \tag{16.12}$$

Suppose we solve an approximation thereto where $u_h \in S_h \not\subset V$, such that

$$a_h(u_h,v_h) = l_h(v_h), \qquad \forall v_h\in S_h,\ h>0. \tag{16.13}$$

We assume that the bilinear form in (16.13) is defined on the linear hull of V and S_h, i.e.

$$a_h : (S_h+V)\times(S_h+V)\to\mathbb{R},$$

and we assume that likewise the linear functional l is well defined on S_h+V. Because $S_h\not\subset V$ we define a mesh-dependent norm to account for the possibility that S_h may not be a subset of H either. We assume, however, that, given a triangulation $\mathcal{T}_h = \{T\}$ of the domain of interest, on each triangle $T\in\mathcal{T}_h$, we still have $v_{|T}\in H$ for all $v\in S_h$. (This will almost always be the case, certainly if we use a polynomial local basis, since H is typically a Sobolev norm like H^1.) We set

$$\|v\|_{\mathcal{T}_h}^2 := \sum_{T\in\mathcal{T}_h}\|v\|_H^2 \tag{16.14}$$

This is a reasonable choice, because it is "backwards" compatible, in that it reduces to $\|v\|_H$ if $v\in H$ globally.

Theorem 16.2 *Let u solve* (16.12)*, and let u_h solve* (16.13)*. Assume there is a constant $\alpha > 0$, independent of h, such that*

$$a_h(v_h, v_h) \geq \alpha \|v_h\|_{\mathcal{T}_h}^2 \qquad \forall v_h \in S_h,\ h > 0.$$

Furthermore, we assume there is a constant $C > 0$, such that

$$a_h(u, v_h) \leq C \|u\|_{\mathcal{T}_h} \|v_h\|_{\mathcal{T}_h} \qquad \forall u \in V + S_h,\ v_h \in S_h,\ h > 0.$$

Then

$$\|u - u_h\|_{\mathcal{T}_h} \leq \left(1 + \frac{C}{\alpha}\right) \inf_{\substack{v_h \in S_h \\ v_h \neq 0}} \|u - v_h\|_{\mathcal{T}_h} + \frac{1}{\alpha} \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|a_h(u, w_h) - l_h(w_h)|}{\|w_h\|_{\mathcal{T}_h}} \tag{16.15}$$

Proof The proof is very similar to the estimates leading to (16.11). We merely have to be make sure to use the mesh-dependent norm (16.14).

Let $v_h \in S_h$ be arbitrary (but such that $u_h \neq v_h$. Otherwise the following inequalities are trivial.) Define $w_h := u_h - v_h$, and consider

$$\begin{aligned}
\alpha \|w_h\|_{\mathcal{T}_h}^2 &\leq a_h(w_h, w_h) \\
&\leq a_h(u_h - v_h, w_h) \pm a_h(u, w_h) \\
&= a_h(u - v_h, w_h) + (l_h(w_h) - a_h(u, w_h)) \\
&\leq C \|u - v_h\|_{\mathcal{T}_h} \|w_h\|_{\mathcal{T}_h} + |l_h(w_h) - a_h(u, w_h)|.
\end{aligned}$$

Dividing by $\|w_h\|_{\mathcal{T}_h}$, we have

$$\begin{aligned}
\alpha \|w_h\|_{\mathcal{T}_h} &\leq C \|u - v_h\|_{\mathcal{T}_h} + \frac{|l_h(w_h) - a_h(u, w_h)|}{\|w_h\|_{\mathcal{T}_h}} \\
&\leq C \|u - v_h\|_{\mathcal{T}_h} + \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|l_h(w_h) - a_h(u, w_h)|}{\|w_h\|_{\mathcal{T}_h}},
\end{aligned}$$

where we have taken the supremum to make the right-hand side of the inequality independent of w_h. Using the triangle inequality

$$\|u - u_h\|_{\mathcal{T}_h} \leq \|u - v_h\|_{\mathcal{T}_h} + \|v_h - u_h\|_{\mathcal{T}_h}$$

we obtain

$$\|u - u_h\|_{\mathcal{T}_h} \leq \left(1 + \frac{C}{\alpha}\right) \|u - v_h\|_{\mathcal{T}_h} + \frac{1}{\alpha} \sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|l_h(w_h) - a_h(u, w_h)|}{\|w_h\|_{\mathcal{T}_h}}$$

and we obtain the assertion, once we note that the inequality holds for all $v_h \in S_h$. ■

As an example we consider a case where we approximate a curved boundary with a piecewise linear approximation. This approximation is given by the edges of the straight-sided triangles adjacent to the boundary. We have previously argued that such an approximation does not compromise the accuracy for linear elements. Now we are in a position to prove this.

Let us first illustrate how this situation can be described by the theory outlined in this section. For simplicity we assume that we solve the Poisson problem with homogeneous boundary condition on a domain Ω with smooth boundary. Assume we are given a triangulation $\mathcal{T}_h$, and

we incorporate the boundary conditions into the solution space. This can be done by requiring $v(z) = 0$ for all points z on the boundary that are vertices of a triangle. We define

$$S_h := \{v \in C^0(\overline{\Omega}) : v|_T \in \mathcal{P}^1(T)\ \forall T \in \mathcal{T}_h,\ v(z) = 0,\ \text{for all vertices } z \text{ on } \partial\Omega\} \tag{16.16}$$

However, this implies that we do not satisfy the boundary conditions $v_{|\partial\Omega} = 0$ exactly, but rather on the edges connecting two points on the boundary. Consider, for example, the configuration shown in Figure 16.1. There are several ways in which to analyze the situation. We shall imagine

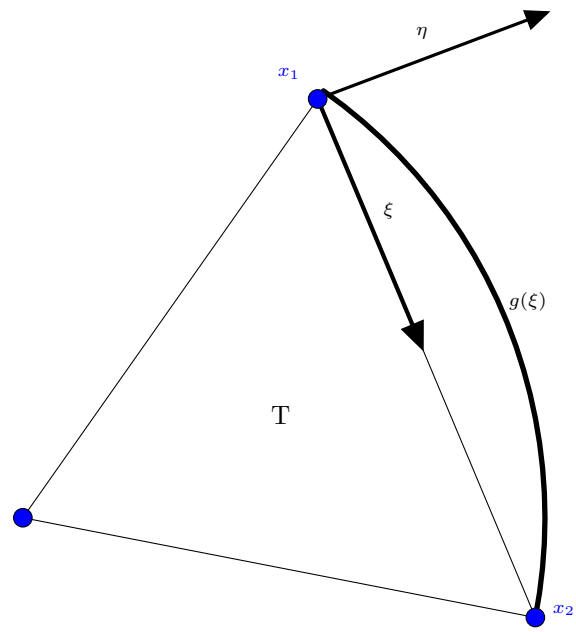

Figure 16.1: A triangle adjacent to the boundary. We assume that we can locally parametrize the boundary using a function g, and local coordinates (ξ, η), as shown in the figure. Note that, according to the definition of the solution space (16.16), we have $v_h = 0$ on the edge connecting x_1 and x_2, but not on the true boundary of the domain.

that we extend the piecewise linear approximate solution to the true boundary $\partial\Omega$. In this case, the numerical solution is defined in all of Ω, as well as on $\partial\Omega$, but evidently $v_{h|\partial\Omega} \neq 0$, and hence $S_h \not\subset H_0^1(\Omega)$. Note, however, that $S_h \subset H^1(\Omega)$, and so $\|v\|_{\mathcal{T}_h} = \|v\|_{1,\Omega}$.

We first need a result concerning the integral along the boundary:

Theorem 16.3 *Let Ω be a domain with smooth boundary, and let $\mathcal{T}_h$ be a quasi-uniform triangulation. Let S_h be defined as in* (16.16)*. Then there is a constant $c > 0$ such that*

$$\|v\|_{0,\partial\Omega} \leq ch^{3/2}|v|_{1,\Omega}, \quad \forall v \in S_h,$$

where $h = \max_{T\in\mathcal{T}_h} h_T$, and h_T is the diameter of triangle T.

Proof Let T be an element adjacent to the boundary, as shown in Figure 16.1. Let us denote as $\Gamma \subset \partial\Omega$ the boundary segment that, using the local coordinate ξ, we have parametrized as $g(\xi)$. We show

$$\int_\Gamma v^2\, ds \leq Ch_T^3 \int_T |\nabla v|^2\, dx \tag{16.17}$$

The result is then obtained by summing over elements. If T is such that all three vertices lie on a boundary, the inequality is trivial ($v = 0$ in this case). Hence we can assume that at most two vertices of the triangle lie on the boundary. By Taylor's series one sees that

$$|g(\xi)| \leq ch_T^2, \qquad \xi_1 \leq \xi \leq \xi_2.$$

where ξ_1 and ξ_2 are the coordinates of the vertices in terms of the local variable ξ. Since v is linear in T, and identically zero on the edge connecting two boundary points, it must have the form

$$v(\xi, \eta) = a\eta, \tag{16.18}$$

for some coefficient a. Since the gradient is constant in each element, we must have $|\nabla v| = a$. We can bound the area of a triangle from below by the area of the incircle. For a quasi-uniform triangulation, as introduced in Definition 13.2, the diameter of the incircle can be bounded from below by the diameter of the triangle. In short, if ρ_T is the diameter of the incircle, then

$$|T| \geq \frac{\pi \rho_T^2}{4} \geq \frac{\pi h_T^2}{4\kappa^2},$$

where κ is the constant appearing in Definition 13.2. Hence we can bound the integral on the right-hand side of (16.17)

$$\int_T |\nabla v|^2 \, dx = a^2 |T| \geq \frac{\pi h_T^2}{4\kappa} a^2. \tag{16.19}$$

On the other hand, using (16.18), and noting $\eta = g(\xi)$ on Γ, we can bound the left-hand side of (16.17), using standard parametrization of the line integral,

$$\begin{aligned} \int_\Gamma v^2 \, ds &= \int_{\xi_1}^{\xi_2} (ag(\xi))^2 \sqrt{1 + g'(\xi)^2} \, d\xi \\ &\leq C(ah_T^2)^2 \int_{\xi_1}^{\xi_2} \sqrt{1 + g'(\xi)^2} \, d\xi \\ &\leq \hat{C}(ah_T^2)^2 h_T \end{aligned} \tag{16.20}$$

Combining (16.19) and (16.20), the statement (16.17) follows. ■

Consider now the problem

$$\begin{aligned} -\nabla \cdot (A\nabla u) &= f, \qquad x \in \Omega, \\ u &= 0, \qquad x \in \partial\Omega. \end{aligned}$$

Integrating by parts, we have

$$\int_\Omega (\nabla v)^T A \nabla u \, dx - \int_{\partial\Omega} \nu \cdot (A\nabla u) \, v \, ds = \int_\Omega f v \, dx. \tag{16.21}$$

Defining $a(u, v) = \int_\Omega (\nabla v)^T A \nabla u \, dx$, and $l(v) := \int_\Omega f v \, dx$, we obtain the usual weak solution as $u \in H_0^1(\Omega)$, such that

$$a(u, v) = l(v) \qquad \forall v \in H_0^1(\Omega)$$

(The boundary term cancels, since v vanishes on the boundary.) If we let $v_h \in S_h$, where V_h is defined in (16.16), one has from (16.21),

$$a(u, v_h) - \int_{\partial\Omega} \nu \cdot (A\nabla u) \, v_h \, ds = l(v_h)$$

As long as $V_h \subset H_0^1(\Omega)$ we can cancel the boundary term, and obtain $a(u, v_h) = l(v_h)$. With the discrete space defined in (16.16), however, this is no longer true. If we assume that the

integration is exact, we can set $a_h(\cdot,\cdot) = a(\cdot,\cdot)$ in (16.15), and likewise $l(\cdot) = l_h(\cdot)$. Thus the boundary integral is precisely the error term $|a(u, v_h) - l(v_h)|$ in (16.15) that is incurred if v_h does *not* satisfy the boundary conditions. We can estimate, using Theorem 16.3 and the trace theorem[1], assuming that $u \in H^2(\Omega)$,

$$\begin{aligned}
\int_{\partial\Omega} \nu \cdot (A\nabla u)\, v_h \, ds &\leq c_A \left\|\nabla u\right\|_{0,\partial\Omega} \left\|v_h\right\|_{0,\partial\Omega} \\
&\leq c_A(\sqrt{|u|_{1,\Omega}^2 + |u|_{2,\Omega}^2})ch^{3/2}|v_h|_{1,\Omega} \\
&\leq \widetilde{C}(\sqrt{|u|_{1,\Omega}^2 + |u|_{2,\Omega}^2})h^{3/2} \left\|v_h\right\|_{1,\Omega}
\end{aligned}$$

Recall that $\|v\|_{\mathcal{T}_h} = \|v\|_{1,\Omega}$ for $v \in V_h$. We see that

$$\sup_{\substack{w_h \in S_h \\ w_h \neq 0}} \frac{|a(u, w_h) - l(w_h)|}{\|w_h\|_{\mathcal{T}_h}} \leq \widetilde{C} h^{3/2} \sqrt{|u|_{1,\Omega}^2 + |u|_{2,\Omega}^2}$$

From Theorem 16.2 and Theorem 13.4 we obtain

$$\|u - u_h\|_{1,\Omega} \leq C \left(h|u|_{2,\Omega} + h^{3/2} \sqrt{|u|_{1,\Omega}^2 + |u|_{2,\Omega}^2} \right)$$

16.1 Further Reading

Our exposition is similar to that in Braess [3].

[1] We use the trace theorem $\|w\|_{0,\partial\Omega} \leq C \|w\|_{1,\Omega}$, and apply it to $w = u_{x_1}$ and $w = u_{x_2}$. Note that $\|\nabla u\|_{1,\Omega}^2 = \|\nabla u\|_{0,\Omega}^2 + |\nabla u|_{1,\Omega}^2 = |u|_{1,\Omega}^2 + |u|_{2,\Omega}^2$.